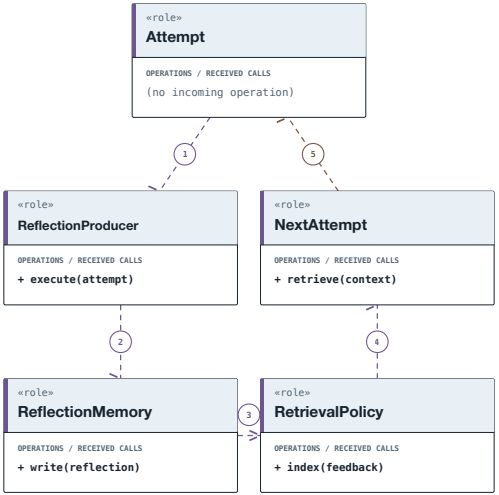
R02 Verbal reflection memory

Reasoning / UML CLASS VIEW / ANALYTICAL MODEL

Reuse feedback across attempts.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	An agent receives feedback that can inform later attempts.
TRIGGER	A failure or critique is verbalized and stored.
STRUCTURE	Reflection producer, retrieval key, memory scope, and update policy.
EFFECT	A later attempt retrieves attributable feedback and uses it in context.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Attempt and feedback identities are linked.
ENSURES	Retrieved reflection retains source and remains updateable or revocable.
INVARIANT	Reflected explanations are not promoted to verified facts.
VIOLATION	Contradictory or poisoned reflections are filtered or explicit.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 Attempt → ReflectionProducer execute(attempt)
2 ReflectionProducer → ReflectionMemory write(reflection)
3 ReflectionMemory → RetrievalPolicy index(feedback)
4 RetrievalPolicy → NextAttempt retrieve(context)
5 NextAttempt ↩ Attempt retry(failure)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Reflexion, episodic memory, observational memory, and project notes.
TEST_OBLIGATION	Test reuse, stale memory, contradiction, poisoning, and deletion.
NEGATIVE_CONTROL	A generic conversation history is not a reflection-memory contract.